Tip: the journey bar is clickable - jump back to any finished stage, or between your tests, anytime.

RESULTS

Results

01 · PAIRED T-TEST

do two related measurements differ?

Table 1. Paired descriptives

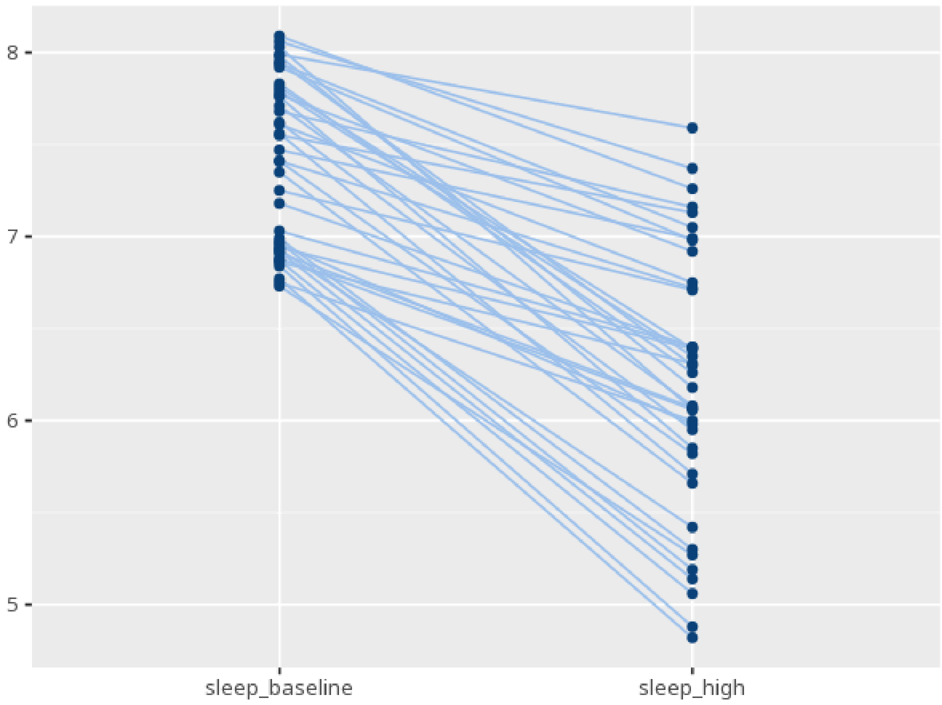
Condition	N	M	SD
sleep_baseline	40	7.40	0.45
sleep_high	40	6.20	0.72

Table 2. Paired-samples t-test

Pair	t	df	p	M_{diff}	95% CI	d [95% CI]$_z$	r (paired)
sleep_baseline – sleep_high	14.14	39	<.001	1.20	[1.03, 1.38]	2.24 [1.65, 2.81]	0.66

assumption check: normality of the difference scores. (Shapiro-Wilk W=0.88, p=<.001) - normality of the differences looks doubtful; consider the Wilcoxon signed-rank test or interpreting with caution

Figure. Change per case



Understanding the numbers

CI. The range of plausible values for the average change between the two conditions. Here, the CI for the mean difference is [1.03, 1.38].

Condition. Which of the two paired conditions this row describes. Here, the two conditions compared are sleep_baseline and sleep_high.

Cohen's dz. The size of the average change in standard-deviation units, for paired data. Here, dz = 2.24 [1.65, 2.81].

df. The degrees of freedom for the paired t-test, based on the number of pairs. Here, df = 39.

M diff. The average change between the two conditions (Condition A – Condition B). Here, M diff = 1.20.

M. The average value recorded in each condition. Here, the mean for sleep_baseline is 7.40 and for sleep_high is 6.20.

N. The number of complete pairs used in the test. Here, N = 40 pairs.

p. The probability of an average change this large (or larger) if there truly were no change. Here, $p < .001$ - below the conventional .05 threshold, a significant change.

SD. How spread out the values are within each condition. Here, the SD for sleep_baseline is 0.45 and for sleep_high is 0.72.

t. How many standard errors the average change is from zero. Here, $t(39) = 14.14$.

r (paired). The Pearson correlation between the two conditions, across cases - how consistently a case that scores high on one also scores high on the other. Here, $r = 0.66$.

How to read this test

Tests whether the average change between two related measurements (e.g. before vs. after) differs from zero. A p below alpha means a significant change; the mean difference, CI and dz describe its size (their sign depends on the subtraction order, A–B). A p at or above alpha does not prove there was no change - only that none was detected. Your significance threshold (α) is 0.05.

APA template: A paired-samples t-test gave M=1.2, $t(39)=14.14$, $p < .001$, dz=2.24 [1.65, 2.81].

Statistical basis: Paired-samples t-test (Student (1908). "The probable error of a mean." Biometrika, 6(1), 1-25.); Cohen's dz effect size (Cohen, J. (1988). Statistical Power Analysis for the Behavioral Sciences (2nd ed.). Routledge.); Paired correlation (Pearson r) between the two conditions (Pearson, K. (1895). "Note on regression and inheritance in the case of two parents." Proceedings of the Royal Society of London, 58, 240-242.). Full references in the exported CITATIONS.txt.